

Solve each equation. Show your work in the space provided.

1. $-5(7 - 2y) + 3(4y + 1) = 12$

$$\begin{aligned} -35 + 10y + 12y + 3 &= 12 \\ 22y - 32 &= 12 \\ 22y &= 44 \\ y &= 2 \end{aligned}$$

2. $6.1(14 - 3h) = -7.9h + 59.4$

$$\begin{aligned} 85.4 - 18.3h &= -7.9h + 59.4 \\ 26 &= 10.4h \\ 2.5 &= h \end{aligned}$$

3. $\frac{7}{5}(-2 + 8k) = \frac{3}{5}(2k + 22)$

$$\begin{aligned} -\frac{14}{5} + \frac{56}{5}k &= \frac{6}{5}k + \frac{66}{5} \\ \frac{50}{5}k &= \frac{80}{5} \\ 10k &= 16 \\ k &= 1.6 \end{aligned}$$

4. $6(4r + 3) = 8(2 + 3r) + 2$

$$\begin{aligned} 24r + 18 &= 16 + 24r + 2 \\ 24r + 18 &= 24r + 18 \\ \text{infinite solutions} \end{aligned}$$

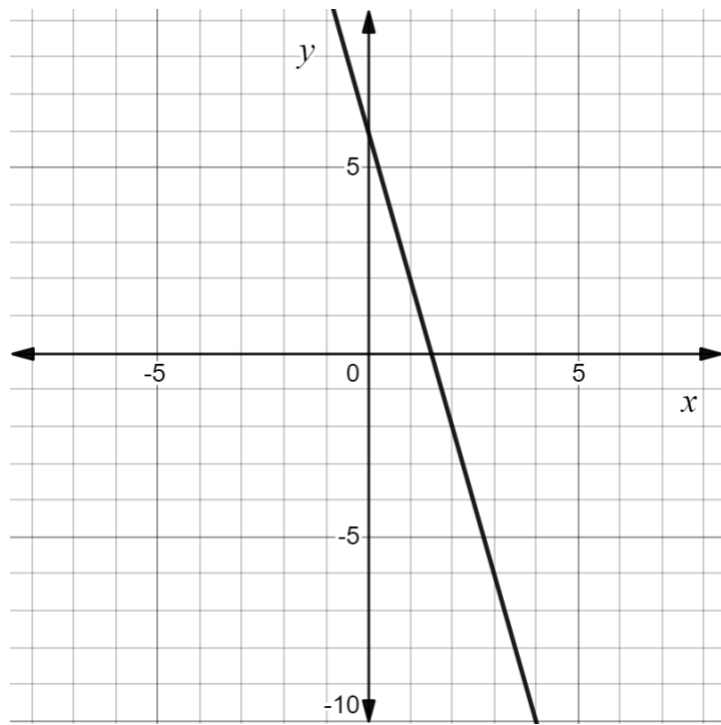
5. $\frac{1}{3}(2x - 54) + 4\frac{1}{3}x = -7$

$$\begin{aligned} \frac{2}{3}x - 18 + 4\frac{1}{3}x &= -7 \\ 5x &= 11 \\ x &= \frac{11}{5} \end{aligned}$$

6. The line $y = -4x + 6$ is shown on the coordinate grid.

Draw a line on the coordinate grid so that the solution to the system of equations graphed on the grid would be "no solution".

Answers will vary. Any line except for the original line, with a slope of -4 is acceptable.



7. A system of equations is shown.

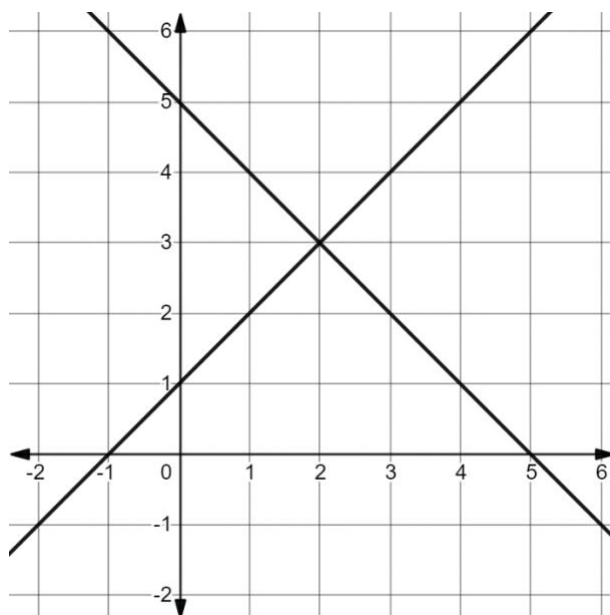
$$\begin{cases} y = -\frac{9}{4}x + 2 \\ y = 3x - 19 \end{cases}$$

Determine which, if any, of the ordered pairs are solutions to the system. Show your work in the provided space.

Ordered Pair	Work	Solution?
a. (8,5)	$5 = -\frac{9}{4}(8) + 2$ $5 = -18 + 2$ $5 = -16$ $5 = 3(8) - 19$ $5 = 24 - 19$ $5 = 5$	☐ Yes ☒ No
b. (4, -7)	$-7 = -\frac{9}{4}(4) + 2$ $-7 = -9 + 2$ $-7 = -7$ $-7 = 3(4) - 19$ $-7 = 12 - 19$ $-7 = -7$	☒ Yes ☐ No

c. $(-4, 11)$	$11 = -\frac{9}{4}(-4) + 2$ $11 = 9 + 2$ $11 = 11$ $11 = 3(-4) - 19$ $11 = -12 - 19$ $11 = -31$	☐ Yes ☒ No
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8. The graph of the system of equations $y = -x + 5$
 $y = x + 1$ is shown.



Complete the statement.

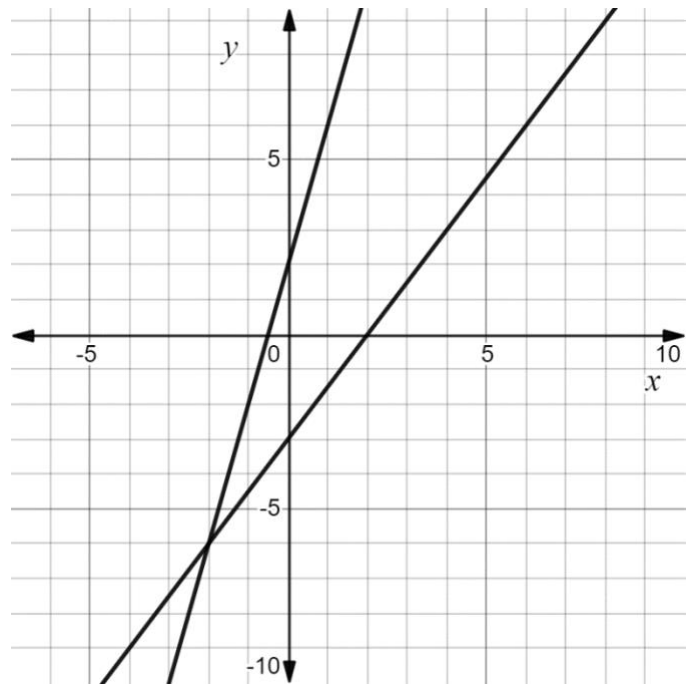
The point $(2, 3)$ is

- ☐ the slope
- ☒ a solution
- ☐ the y -intercept

of

- ☐ $y = x + 1$ only.
- ☐ $y = -x + 5$ only.
- ☒ $y = -x + 5$ and $y = x + 1$.

9. The graph of the system $\begin{cases} y = \frac{3}{2}x - 3 \\ y = 4x + 2 \end{cases}$ is shown.

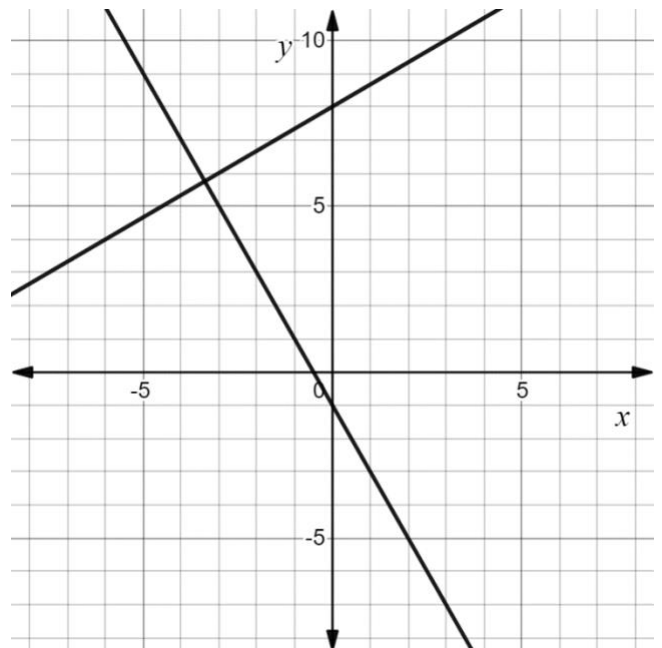


What is the solution to this system?

Solution

$(-2, -6)$

10. The graph of the system $\begin{cases} y = -2x - 1 \\ y = \frac{2}{3}x + 8 \end{cases}$ is shown.



Answers may vary. The exact solution is $(-3.375, 5.75)$, but students are only expected to approximate from the graph. An example approximation students might give is shown. Reasonable approximations based on the graph should be accepted.

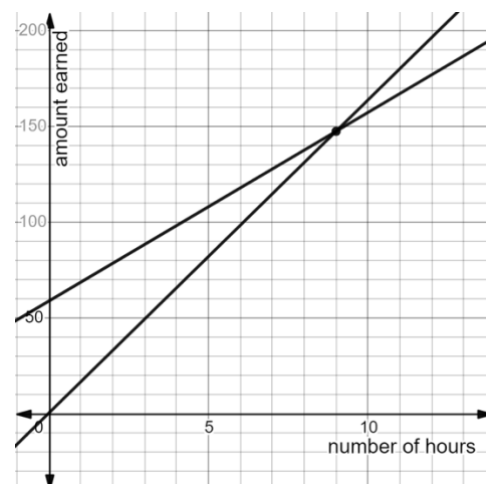
What is the approximate solution to this system?

**Approximate
Solution**

$(-3.5, 5.75)$

11. Serena just started working at the local salon. She is paid \$9.85 per hour and earns, on average, \$58.50 in tips each day. Roseline just started working at a bank. Since she does not receive tips, she earns \$16.35 an hour. The system of equations $\begin{cases} y = 9.85x + 58.50 \\ y = 16.35x \end{cases}$ can be used to represent the amount of money earned, y , each day for working x hours.

The graph of the system is shown.

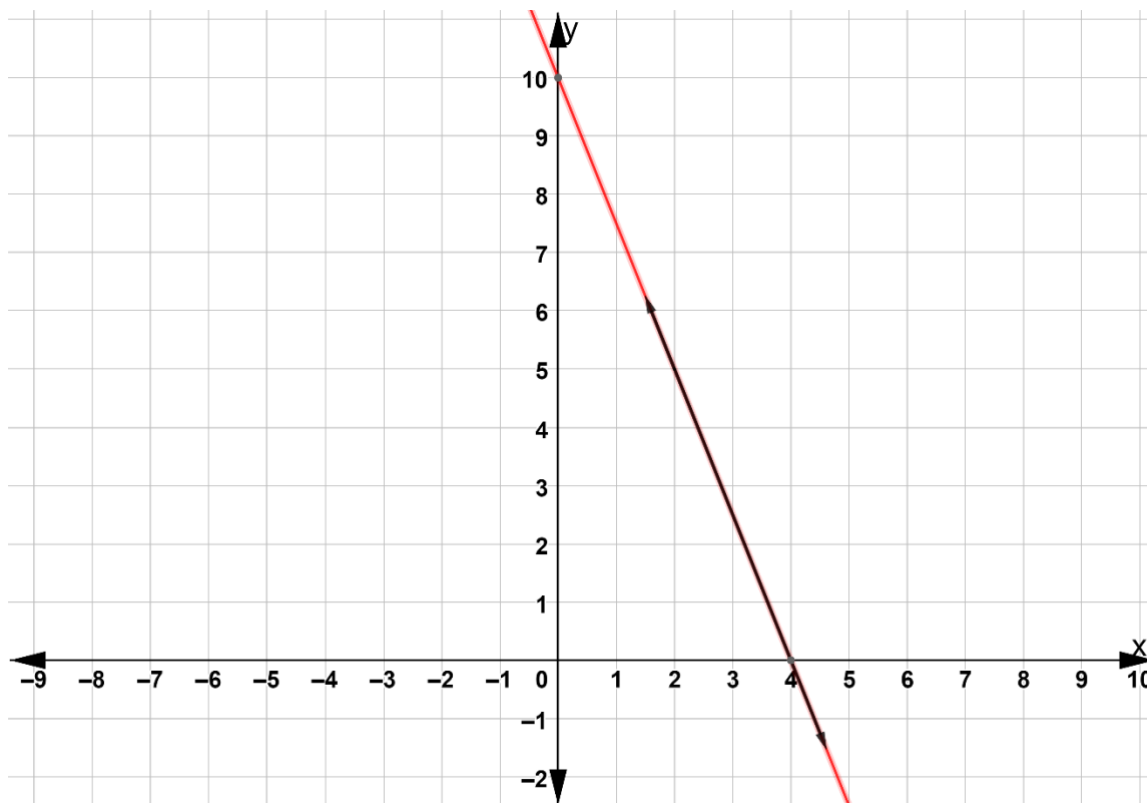


What is the approximate solution to this system?

Approximate Solution	(9, 147.15)
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12. The equation $y = 10 - 2x$ represents the amount of water, y , in gallons in Tank 1 after x seconds. The equation $6x + 3y = 30$ represents the amount of water, y , in gallons in Tank 2 after x seconds.

A coordinate grid is shown, where the equation $y = 10 - 2.5x$ is graphed.



- a. Graph $y = -\frac{5}{2}x + 10$ to complete the graph of the system $\begin{cases} y = 10 - 2.5x \\ y = -\frac{5}{2}x + 10 \end{cases}$.

- b. Complete the statements.

The two equations in the system are lines with

- ☒ the same slope.
- ☐ different slopes.

The system $\begin{cases} y = 10 - 2.5x \\ y = -\frac{5}{2}x + 10 \end{cases}$

- ☐ has no solution.
- ☐ has one solution.
- ☒ has an infinite number of solutions.